UNIVERSITY INSTITUTE of COMPUTING
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NAAC GRADE A+ ACCREDITED UNIVERSITY

EXPERIMENT- 1

AIM:

Formatting the worksheet using logical formula IF Statement, Nested IF, AND, OR, NOT, IFERROR, SUMIF, AVERAGEIF, COUNTIF and AVERAGEIF)

OBJECTIVE:

1. To understand and apply logical functions in Microsoft Excel.
2. To determine student eligibility based on attendance using the **IF** function.
3. To evaluate pass/fail status using the **AND** function.
4. To identify distinction holders using the **OR** function.
5. To handle formula errors gracefully using the **IFERROR** function.
6. To calculate total marks using the **SUMIF** function.
7. To calculate average marks using the **AVERAGEIF** function.
8. To count the number of students meeting specific criteria using the **COUNTIF** function.

PROCEDURE:

1. Open Microsoft Excel and create a new worksheet.
2. Enter the student data (Attendance and marks of 5 subjects).
3. Format the worksheet with headings and borders.
4. Apply the required formulas (IF, Nested IF, AND, OR, NOT, IFERROR, SUMIF, AVERAGEIF, and COUNTIF).
5. Verify the calculated results.
6. Save the worksheet.

IF FORMULA:

Definition:

The IF function checks whether a condition is TRUE or FALSE and returns one value if TRUE and another value if FALSE.

Condition:

Attendance is greater than 75%.

Formula:

=IF(C2>75,"Eligible","Not Eligible")

Output:

I2 | =IF(C2>75,"Eligible","Not Eligible")

	A	B	C	D	E	F	G	H	I
1	Roll No	Student Name	Attendance %	Mathematics	English	Computer	Science	Hindi	IF FORMULA
2	1	Aarav Sharma	82	85	78	92	88	81	Eligible
3	2	Aditi Verma	69	62	58	71	65	60	Not Eligible
4	3	Aman Singh	91	95	90	93	97	94	Eligible
5	4	Ananya Gupta	76	74	72	69	71	75	Eligible
6	5	Arjun Kumar	58	45	39	52	48	55	Not Eligible
7	6	Bhavya Kaur	88	91	84	88	90	93	Eligible
8	7	Deepak Yadav	72	68	73	70	75	71	Not Eligible
9	8	Diya Sharma	95	98	96	94	99	97	Eligible

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AND FORMULA:

Definition:

The AND function returns TRUE only if all specified conditions are TRUE.

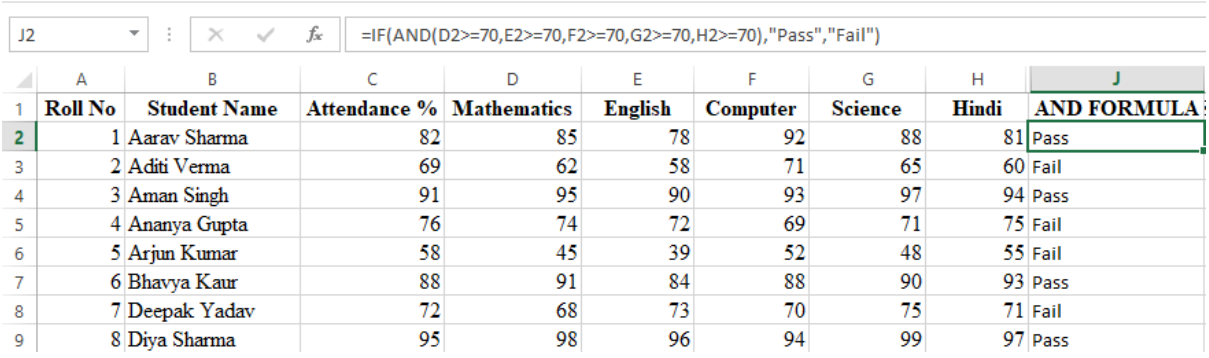
Condition:

All 5 subjects must have marks 70 or above.

Formula:

=IF(AND(D2>=70,E2>=70,F2>=70,G2>=70,H2>=70),"Pass","Fail")

Output:



Roll No	Student Name	Attendance %	Mathematics	English	Computer	Science	Hindi	AND FORMULA
1	Aarav Sharma	82	85	78	92	88	81	Pass
2	Aditi Verma	69	62	58	71	65	60	Fail
3	Aman Singh	91	95	90	93	97	94	Pass
4	Ananya Gupta	76	74	72	69	71	75	Fail
5	Arjun Kumar	58	45	39	52	48	55	Fail
6	Bhavya Kaur	88	91	84	88	90	93	Pass
7	Deepak Yadav	72	68	73	70	75	71	Fail
8	Diya Sharma	95	98	96	94	99	97	Pass

SUMIF FORMULA:

Definition:

The SUMIF function adds only those values that satisfy a specified condition.

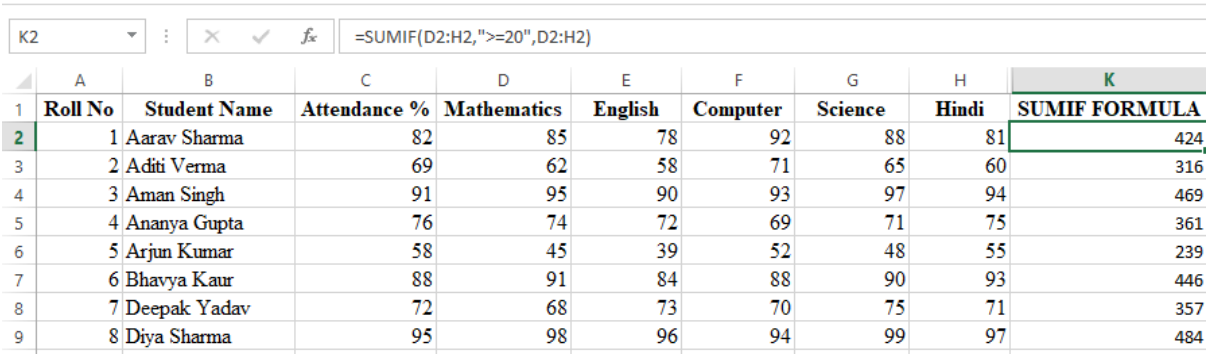
Condition:

Add marks that are 20 or above**.**

Formula:

=SUMIF(D2:H2,">=20",D2:H2)

Output:



Roll No	Student Name	Attendance %	Mathematics	English	Computer	Science	Hindi	SUMIF FORMULA
1	Aarav Sharma	82	85	78	92	88	81	424
2	Aditi Verma	69	62	58	71	65	60	316
3	Aman Singh	91	95	90	93	97	94	469
4	Ananya Gupta	76	74	72	69	71	75	361
5	Arjun Kumar	58	45	39	52	48	55	239
6	Bhavya Kaur	88	91	84	88	90	93	446
7	Deepak Yadav	72	68	73	70	75	71	357
8	Diya Sharma	95	98	96	94	99	97	484

AVERAGEIF FORMULA:

Definition:

The AVERAGEIF function calculates the average of values that meet a specified condition.

Condition:

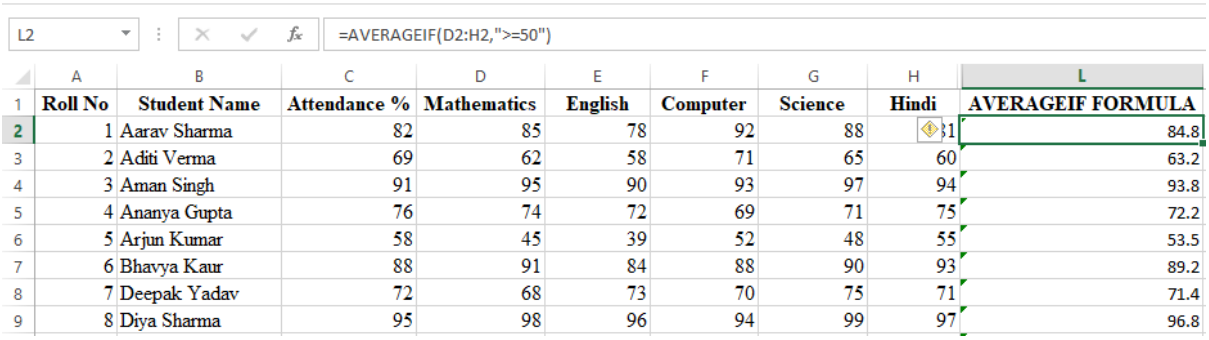
Calculate the average of subject marks 50 or above**.**

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Formula:

=AVERAGEIF(D2:H2,">=50")

Output:



	A	B	C	D	E	F	G	H	L
1	Roll No	Student Name	Attendance %	Mathematics	English	Computer	Science	Hindi	AVERAGEIF FORMULA
2	1	Aarav Sharma	82	85	78	92	88	81	84.8
3	2	Aditi Verma	69	62	58	71	65	60	63.2
4	3	Aman Singh	91	95	90	93	97	94	93.8
5	4	Ananya Gupta	76	74	72	69	71	75	72.2
6	5	Arjun Kumar	58	45	39	52	48	55	53.5
7	6	Bhavya Kaur	88	91	84	88	90	93	89.2
8	7	Deepak Yadav	72	68	73	70	75	71	71.4
9	8	Diya Sharma	95	98	96	94	99	97	96.8

NESTEDIF FORMULA:

Definition:

A Nested IF function contains multiple IF statements to evaluate several conditions in sequence.

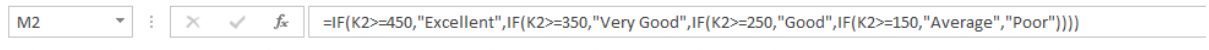
Condition:

Performance is determined based on Total Marks (SUMIF result).
450 or above → Excellent
350–449 → Very Good
250–349 → Good
150–249 → Average
Below 150 → Poor

Formula:

=IF(K2>=450,"Excellent",IF(K2>=350,"Very Good", IF(K2>=250,"Good", IF(K2>=150,"Average","Poor"))))

Output:



	A	B	D	E	F	G	H	L	M
1	Roll No	Student Name	Mathematics	English	Computer	Science	Hindi	AVERAGEIF FORMULA	NESTEDIF FORMULA
2	1	Aarav Sharma	85	78	92	88	81	84.8	Very Good
3	2	Aditi Verma	62	58	71	65	60	63.2	Good
4	3	Aman Singh	95	90	93	97	94	93.8	Excellent
5	4	Ananya Gupta	74	72	69	71	75	72.2	Very Good
6	5	Arjun Kumar	45	39	52	48	55	53.5	Average
7	6	Bhavya Kaur	91	84	88	90	93	89.2	Very Good
8	7	Deepak Yadav	68	73	70	75	71	71.4	Very Good
9	8	Diya Sharma	98	96	94	99	97	96.8	Excellent

OR FORMULA:

Definition:

The OR function returns TRUE if at least one of the specified conditions is TRUE.

Condition:

If any one of the 5 subjects has marks 90 or above, the student gets Distinction.

Formula:

=IF(OR(D2>=90,E2>=90,F2>=90,G2>=90,H2>=90),"Distinction","No Distinction")

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Output:

N2 | =IF(OR(D2>=90,E2>=90,F2>=90,G2>=90,H2>=90),"Distinction","No Distinction")

Roll No	Student Name	Attendance %	Mathematics	English	Computer	Science	Hindi	OR FORMULA
1	Aarav Sharma	82	85	78	92	88	81	Distinction
2	Aditi Verma	69	62	58	71	65	60	No Distinction
3	Aman Singh	91	95	90	93	97	94	Distinction
4	Ananya Gupta	76	74	72	69	71	75	No Distinction
5	Arjun Kumar	58	45	39	52	48	55	No Distinction
6	Bhavya Kaur	88	91	84	88	90	93	Distinction
7	Deepak Yadav	72	68	73	70	75	71	No Distinction
8	Diya Sharma	95	98	96	94	99	97	Distinction

ERRORIF FORMULA:

Definition:

The IFERROR function returns a specified value if a formula results in an error; otherwise, it returns the formula result.

Condition:

If the AVERAGEIF result in column **L** contains an error (such as #DIV/0!), display "ERROR"; otherwise, display the average value.

Formula:

=IFERROR(L2,"ERROR")

Output:

O2 | =IFERROR(L2,"ERROR")

Roll No	Student Name	Mathematics	English	Computer	Science	Hindi	AVERAGEIF FORMULA	ERRORIF FORMULA
1	Aarav Sharma	85	78	92	88	81	84.8	84.8
2	Aditi Verma	62	58	71	65	60	63.2	63.2
3	Aman Singh	95	90	93	97	94	93.8	93.8
4	Ananya Gupta	74	72	69	71	75	72.2	72.2
5	Arjun Kumar	45	39	52	48	55	53.5	53.5
6	Bhavya Kaur	91	84	88	90	93	89.2	89.2
7	Deepak Yadav	68	73	70	75	71	71.4	71.4
8	Diya Sharma	98	96	94	99	97	96.8	96.8
9	Harsh Mehta	56	61	64	59	63	60.6	60.6
10	Isha Kapoor	79	81	85	80	84	81.8	81.8
11	Jai Patel	40	45	50	35	42	50	50
12	Kanika Jain	88	91	86	90	89	88.8	88.8
13	Karan Gill	72	74	76	78	80	76	76
14	Khushi Arora	94	89	91	93	95	92.4	92.4
15	Manav Saini	67	69	71	72	74	70.6	70.6
16	Mehak Bansal	82	85	88	84	87	85.2	85.2
17	Mohit Kumar	30	42	38	46	40	#DIV/0!	ERROR
18	Muskan Sharma	96	94	92	95	97	94.8	94.8

COUNTIF FORMULA:

Definition:

The COUNTIF function counts the number of cells that meet a specified condition.

Condition:

Count the number of students with "Distinction".

Formula:

=COUNTIF(N2:N51,"Distinction")

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Output:

P2 | =COUNTIF(N2:N51,"Distinction")

	A	B	D	E	F	G	H	N	P
1	Roll No	Student Name	Mathematics	English	Computer	Science	Hindi	OR FORMULA	COUNTIF FORMULA
2	1	Aarav Sharma	85	78	92	88	81	Distinction	21
3	2	Aditi Verma	62	58	71	65	60	No Distinction	
4	3	Aman Singh	95	90	93	97	94	Distinction	
5	4	Ananya Gupta	74	72	69	71	75	No Distinction	
6	5	Arjun Kumar	45	39	52	48	55	No Distinction	
7	6	Bhavya Kaur	91	84	88	90	93	Distinction	
8	7	Deepak Yadav	68	73	70	75	71	No Distinction	
9	8	Diya Sharma	98	96	94	99	97	Distinction	

LEARNING OUTCOMES:

1. Understand the use of basic Excel formulas.
2. Learn how to use logical functions in Excel.
3. Apply formulas to calculate marks and attendance.
4. Check student eligibility and performance using formulas.
5. Create and format a worksheet in Microsoft Excel.

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